

# Screens

Wander into the realm of displays that show you the world on your desktop

Everyone wants to enjoy breathtaking and dazzling graphics of their favourite movies on their PC monitor. Entertainment and multimedia are driving forces that are changing the capabilities of the humble computer display.

The market for Cathode Ray Tube (CRT) monitors and Liquid Crystal Displays (LCDs) is on the rise, with manufacturers tweaking the interface for more comfort and user friendliness. With big brands falling all over themselves to the consumer's heart and wallet, soon you may have to make the choice between a TV and a computer. And with the new range of monitors coming with remote controls, the TV is surely on its way out. And not only is it vying for the space occupied by the TV, with its USB ports, the new LCDs and CRTs are also showing the Mac the door.

## How we tested

The monitors were tested keeping in mind the performance, features, value for money and the warranty offered.

**Performance:** We used the Display Mate benchmarking software to gauge a monitor's performance. This software uses images to test a monitor for criteria such as geometrical accuracy seen in the instance of parallel lines. This aspect is especially important in applications where accurate geometric shapes need to be generated like in CAD and CAM software.

Another important factor to consider is the ability of the monitor to hold a stable shape of images. Sudden changes in brightness levels occur—like when you are watching a movie—can create distortion. This leads to eyestrain and is caused when the monitor lacks the precision to prevent adjacent pixels from lighting up.

In the sharpness and resolution set of tests, we tested the image sharpness offered by the monitor and the resolution it displayed pictures clearly at. Another check point was if, while watching a

movie or a moving object, the screen displayed the object with same sharpness across the screen, as it did in the centre.

We also evaluated the ability of a monitor to generate perfect lines (including vertical, horizontal and diagonal), circles and various other geometric patterns. Even the highest resolution systems show jagged lines, inability to discriminate between closely spaced lines and object fusion or merging of pixels.

The colour and grayscale test helped gauge how well the monitor is able to reproduce colour, as well as any type of ghosting and streaking that might happen with different colour shades, brightness and contrast. It also checked for overlapping of darker or lighter shades on other colours. This test had the highest number of sub-tests (nine) as well as the highest weightage.

A few tests other effects such as the uniformity of the display and a flicker test were also carried out. In the flicker test, a white background was displayed by the test utility, and the monitor was run at different resolutions to see if it flickers. The Defocusing, Blooming and Halos test, told us how much thicker an object will look than its darker counterpart.

The pixel persistence test was run specifically for the LCDs to check for jerks or image blurring. Another LCD-specific test was the viewable angle test. Unlike CRT monitors, when you view an LCD screen from an angle, things are not as bright and clear as with a straight view. A document containing different fonts was viewed from various angles, thereby letting us determine the angles at which the object still looked crisp and sharp.

**Features:** In the features test we looked for value-additions and other aspects which aimed at increasing the user-friendliness of the display. The features noted varied in their significance—some things as simple as the clarity of the fonts on the OSD (On Screen Display) and the presence of energy saving features that can be enabled from the BIOS. Other